LEARNER GUIDE

Cyclic quadrilaterals

Circle Theorems · Lesson 5. A short guide to walk beside you — read as much or as little as helps. *Connection before curriculum. Always.*

What this lesson is about

A **cyclic quadrilateral** is a four-sided shape whose four vertices all lie on a circle. Measure its angles and a pattern appears: the two pairs of **opposite** angles each add to **180°** — ∠A + ∠C = 180° and ∠B + ∠D = 180°. The opposite angles are **supplementary**. It is the keystone once more: opposite corners look across the two arcs on either side of a diagonal, and the central angles of those two arcs together make a full **360°** turn, so each pair of corner angles (being half of each central angle) adds to half of 360°, which is 180°. A useful extra: the **exterior** angle of a cyclic quadrilateral equals the opposite **interior** angle.

What you are learning to notice

- **Opposite angles of a cyclic quadrilateral add to 180°** (they are supplementary).
- It is the keystone again: each pair is half of a full 360° turn at the centre.
- Use OPPOSITE angles (across the shape), not adjacent ones.
- The exterior angle equals the opposite interior angle.

Moving through it, one step at a time

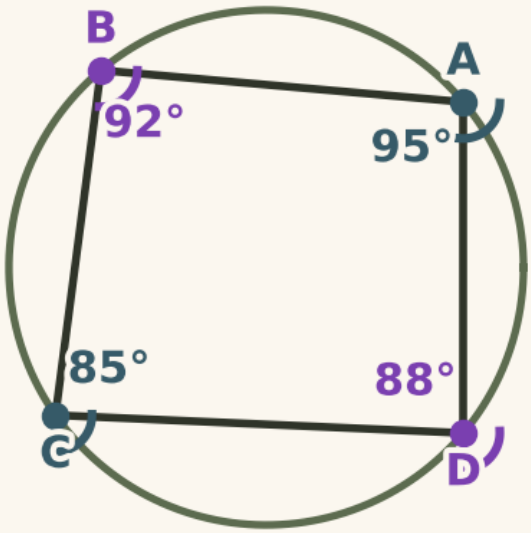
Check the four vertices are on the circle (a cyclic quadrilateral). To find a missing angle, take 180° minus its opposite angle, and give the reason 'opposite angles of a cyclic quadrilateral add to 180°'. If you know three angles, use both opposite pairs to find the fourth.

Worked examples

Two full examples to follow. Read them slowly — the aim is to see how each line follows from the one before, not to memorise a rule. It can help to cover the steps and predict the next line before you reveal it.

Example 1 — Find the opposite angle

ABCD is a cyclic quadrilateral. Reading its angles, ∠A = 95° and ∠B = 92°. Find ∠C and ∠D.



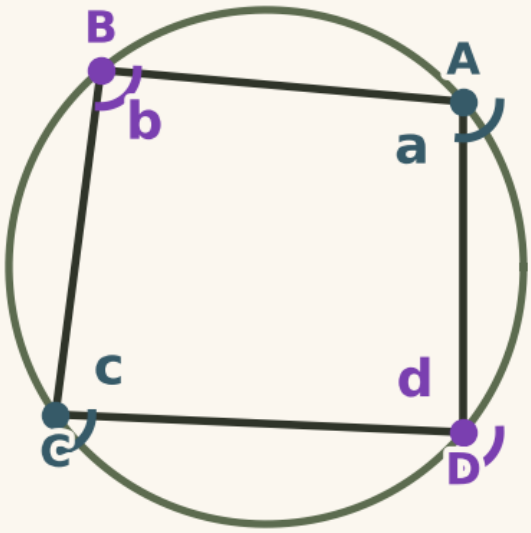
1. Opposite angles add to 180°: ∠A + ∠C = 180°. ∠C = 180 – 95

2. And ∠B + ∠D = 180°. ∠D = 180 – 92

∠C = 85° and ∠D = 88°, because opposite angles of a cyclic quadrilateral add to 180°.

Example 2 — Why they add to 180°

Prove ∠A + ∠C = 180° for a cyclic quadrilateral. Let ∠A = a and ∠C = c.



1. ∠A stands on arc BCD; ∠C stands on arc BAD (the two arcs of diagonal BD).	two arcs
2. Their central angles are 2a and 2c (angle at the centre).	2a and 2c
3. These two central angles make a full turn: 2a + 2c = 360°.	full 360° turn

Dividing by 2: a + c = 180°, i.e. ∠A + ∠C = 180°. Opposite angles of a cyclic quadrilateral are supplementary. ■

A gentle reminder

You can pause. You can return. You do not need to complete everything in one sitting.

If you get stuck

Getting stuck is part of doing mathematics, not a sign that something is wrong. There is support built into the lesson, and using it is a strong move:

- **Reconnection Routes** — if an earlier idea feels shaky (the angle at the centre theorem (Lesson 2), that angles at a point add to 360°, and that the angles of a quadrilateral add to 360°), open a short recap and return whenever you are ready.
- **Socratic prompt** — a question to nudge your thinking.

- **Hint** — a little more help with the step in front of you.
- **Check** — try an answer and see how you are doing; a wrong try is useful information.
- **Worked solution** — a full model to compare your thinking against.

How the worked solution unlocks

The worked solution unlocks after you use the Socratic prompt, use the hint, and check two answers. Those steps are the learning — the worked solution is there to confirm your thinking, not to replace it.

Using support is part of learning. It is not cheating.

Recording your thinking

You can show your working in whatever way suits you today — the on-screen Scratchpad, paper, a spoken explanation, or a photo of a sketch. There is no single right format. If you do not finish everything in one sitting, that is completely fine: what you have done is information for your next step, never a failure.

Before you begin

Settle into a comfortable pace and take the steps one at a time. If your attention drifts, pause and come back — the lesson will keep your place. When you are ready, begin.

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